LATITUDE MEDTECH

AI Multi-Agent Consulting System

Master Execution Build Plan

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North Star	AI-powered medical device consulting platform delivering FDA, EU MDR, ISO 13485, and MDSAP advisory through autonomous agents with voice interaction and human SME oversight.
Starting Wedge	Internal team tool: FDA regulatory agent + document generation + human review queue.
Frameworks	FDA 21 CFR, EU MDR 2017/745, ISO 13485:2016, MDSAP, ISO 14971
Architecture	Claude (Anthropic) + LangGraph orchestration + pgvector RAG + FastAPI + React + Whisper/ElevenLabs voice
Phases	5 phases over ~12 months: Core → RAG → Voice → Full Agent Roster → Client-Facing Hardening

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1. Company & Project Overview

Company Name

- Recommended: Latitude MedTech
- Domain positioning: precision navigation through regulatory complexity
- Tagline candidate: Navigating compliance. Accelerating innovation.

Mission

Build an autonomous, voice-activated AI consulting platform that functions as a full medical device regulatory consulting firm — staffed by specialized AI agents, overseen by human SMEs, and capable of delivering FDA, EU MDR, ISO 13485, and MDSAP advisory to medical device companies at scale.

Market Context

- Medical device regulatory consulting is a $4B+ market with chronic talent shortages and high per-hour consulting costs ($300–600/hr for senior regulatory affairs specialists).
- Small and mid-size device companies are underserved — they lack in-house expertise and cannot afford full-time RA staff.
- AI can compress timelines for submission readiness, gap analysis, and QMS documentation from months to days.
- Human oversight remains non-negotiable for regulated outputs — the system augments, not replaces, licensed consultants.

Users

- Primary (internal): Regulatory affairs consultants using the system to 10x their throughput.
- Secondary (client-facing): Quality and RA managers at medical device manufacturers.
- Oversight: Principal consultants / SMEs who gate all client-deliverable outputs.

2. Orchestration Model & Roles

This system follows the Master Orchestrator doctrine. The AI Orchestrator owns the outcome, delegates to specialized agents, and accepts or rejects all sprint outputs. The human counterpart acts as field agent.

Orchestrator Responsibilities

- Define scope, phases, and acceptance criteria for every sprint.
- Spawn and manage specialized AI workers with disjoint ownership lanes.
- Collect agent reports, run integration, assign independent verifiers.
- Make final ACCEPT / REJECT decisions on all sprint deliverables.
- Maintain command center docs and keep implementation synchronized with docs.
- Escalate blockers requiring human action using the Human Counterpart Task format.

Human Counterpart Role & Boundaries

Authority	Boundaries
Register legal entity (Latitude MedTech LLC/Corp)	Cannot approve AI outputs as regulatory advice without licensed SME review
Procure API keys, cloud accounts, domain	Cannot merge code to production without orchestrator ACCEPT

Conduct client discovery calls	Cannot make regulatory claims to clients from unverified agent outputs
Sign vendor contracts under $5K	Contracts above $5K require explicit orchestrator task assignment + SME review
Test and report bugs in staging	Cannot deploy to production without orchestrator sign-off

3. AI Agent Roster

The system comprises 4 agent tiers. Every agent has a defined owner, scope, and reporting format per the Operating Doctrine.

Tier 1 — Orchestration Layer

Agent	Role	Key Responsibilities
Orchestrator	Master owner	Routes all queries, manages state, accepts/rejects outputs
Program Manager	Client engagement lead	Timelines, deliverable tracking, escalation, client status reports

Tier 2 — Regulatory Agents

Agent	Frameworks Owned	Key Outputs
FDA Agent	21 CFR 820, 510(k), PMA, De Novo, EUA	Submission strategy, predicate analysis, regulatory pathway memos
EU MDR Agent	EU MDR 2017/745, IVDR	Technical files, GSPR checklists, notified body selection guidance
ISO 13485 / MDSAP Agent	ISO 13485:2016, MDSAP audit program	QMS gap analysis, audit readiness reports, procedure templates
Regulatory Strategy Agent	Cross-market	Market entry strategy, parallel submission plans, timeline modeling

Tier 3 — Process Owner Agents (QMS Clause Owners)

Agent	ISO Clause / Standard	Trigger Events
Management Review	ISO 13485 §5.6, MDSAP	Periodic review cycle, CAPA escalation, audit findings
CAPA Agent	ISO 13485 §8.5, 21 CFR 820.100	Nonconformance filed, complaint received, audit finding
Risk Management	ISO 14971:2019	Design change, new hazard identified, CAPA outcome
Design Controls	ISO 13485 §7.3, 21 CFR 820.30	New product, design change request, DHF review
Document Control	ISO 13485 §4.2, 21 CFR 820.40	SOP creation/revision, change control request
Supplier Quality	ISO 13485 §7.4, MDSAP	New supplier, re-qualification, supplier audit finding
Complaint Handling	ISO 13485 §8.2, MDR Article 87	Customer complaint, field report, adverse event
Training Agent	ISO 13485 §6.2	New hire, procedure update, competency gap

Internal Audit	ISO 13485 §8.2.2	Scheduled audit, triggered by CAPA or NC
Production & Process	ISO 13485 §7.5	Process validation request, IQ/OQ/PQ event

Tier 4 — Support & Growth Agents

Agent	Function	Interacts With
Document Generation	Drafts SOPs, reports, submission templates	All process owners, Human Review
Human Review Agent	Gates all client-facing outputs, routes to SME queue	All agents, human SMEs
Disclaimer Agent	Appends regulatory disclaimer to all outputs	Document Generation, Human Review
RAG Knowledge Base	Retrieves relevant guidance docs for all agents	All regulatory + process owner agents
Sales Agent	Pipeline tracking, proposal drafting, CRM updates	Program Manager, Regulatory agents
Marketing Agent	Content creation, campaign briefs, lead nurturing	Document Generation, Sales Agent
Prof. Development Agent	Training plans, cert tracking, skill gap analysis	Training Agent, Internal Audit

4. Agent Interaction & Trigger Chains

Agents interact via three mechanisms: event triggers (automatic), feed relationships (data passing), and escalation paths (human-gated).

Trigger Event	Chain	Human Gate?
Nonconformance filed	CAPA → Risk Management → Management Review	Yes — at Management Review
Design change request	Design Controls → Doc Control → Risk Management	Yes — at Risk Management output
Supplier issue	Supplier Quality → CAPA → Complaint Handling (if field impact)	Yes — at CAPA output
Audit finding	Internal Audit → CAPA → Management Review	Yes — at Management Review
New client lead	Sales Agent → Program Manager → Regulatory Agents	Yes — PM assigns pathway
Content request	Marketing Agent → Document Generation → Human Review	Yes — before external publish
Training gap identified	Prof Dev Agent → Training Agent → Internal Audit (if systemic)	Yes — if systemic finding
Any client-facing output	Any Agent → Document Generation → Human Review → Disclaimer → Delivery	**Always**

5. Technical Architecture

Layer	Technology	Rationale
LLM	Claude (Anthropic API) — claude-sonnet-4-6	Tool use, long context, strong instruction following for regulatory precision
Orchestration	LangGraph (stateful agent graph)	Supports complex multi-agent flows, human-in-the-loop nodes, state persistence
Voice STT	OpenAI Whisper (self-hosted or API)	Accurate medical/regulatory terminology recognition
Voice TTS	ElevenLabs or Azure Cognitive TTS	Natural voice for client-facing interactions
RAG Store	PostgreSQL + pgvector	Scales well, easy ops, supports metadata filtering by framework
RAG Pipeline	LangChain document loaders + chunking	Handles FDA guidance PDFs, ISO standards, internal SOPs
API Layer	FastAPI (Python)	Async, modern, well-documented, easy WebSocket support
Frontend	React + WebSocket (real-time voice)	Real-time streaming responses, voice UI components
Auth	Auth0 or Clerk	Role-based access: internal consultant vs. client vs. SME reviewer
Review Queue	PostgreSQL + FastAPI + React UI	Persistent queue, confidence scoring, SME approval workflow
Logging / Audit	Structured JSON logs → S3/Postgres	Full audit trail of agent decisions — required for regulated context
Infra	AWS (ECS Fargate + RDS + S3)	Managed, HIPAA-eligible infrastructure if patient data ever in scope
Secrets	AWS Secrets Manager	No secrets in code or env files in repos

Data Model Assumptions

- All client data is tenant-isolated by client_id at the database level from day one.
- No PHI (protected health information) will be processed without a formal HIPAA BAA and security review.
- RAG knowledge base uses only public FDA/EU guidance documents and Latitude-owned SOP templates initially.
- Licensed ISO standards (13485, 14971) are NOT ingested verbatim — only Latitude-authored interpretation and procedure templates.
- All agent outputs are logged with agent_id, timestamp, confidence score, and human reviewer ID.
- Outputs are explicitly marked: Demo / Alpha / Beta / Production-ready at all times.

6. Build Phases

Phase 1 — Core Vertical Slice *Weeks 1–6*

- Orchestrator + FDA Agent (single specialist)
- Document Control process owner agent
- Document Generation agent
- Human Review queue (basic)
- Disclaimer agent
- FastAPI skeleton + React chat UI (text only)
- Basic RAG pipeline: 20 FDA guidance docs ingested

Verification Gate: SME can use system to draft a 510(k) strategy memo, route to review queue, approve, and deliver with disclaimer.

Phase 2 — RAG + Full Regulatory Roster *Weeks 7–12*

- EU MDR Agent + ISO/MDSAP Agent + Reg Strategy Agent
- Expand RAG: EU guidance, ISO-aligned templates, internal SOPs
- Confidence scoring on agent outputs
- Review queue hardening: role-based access, SME assignment
- Structured logging + audit trail

Verification Gate: System covers all 4 regulatory frameworks. SME can run a multi-market gap analysis.

Phase 3 — Voice Layer *Weeks 13–16*

- Whisper STT integration
- TTS response synthesis
- WebSocket real-time streaming in React UI
- Voice-activated query routing through Orchestrator
- Voice session logging

Verification Gate: User can speak a regulatory question and receive a voiced, reviewed response.

Phase 4 — Full Process Owner Roster *Weeks 17–24*

- All 10 process owner agents deployed (CAPA, Risk, Design, Supplier, etc.)
- Cross-agent trigger chains implemented in LangGraph
- Program Manager agent: timeline tracking + status reports
- Sales, Marketing, Prof Development agents
- Multi-agent workflow testing

Verification Gate: End-to-end QMS advisory workflows operational. NC → CAPA → Risk → Mgmt Review chain verified.

Phase 5 — Client-Facing Hardening	*Weeks 25–36*

- Auth0 multi-tenant auth with client roles
- Client portal (React): submit questions, view outputs, track open actions
- Rate limiting, cost controls, usage metering
- Security review + penetration test
- Beta client onboarding (3–5 design partners)
- SLA definitions and monitoring

Verification Gate: First paying clients onboarded. System operates at Beta readiness level.

7. Sprint 1 Plan — Core Vertical Slice

Duration: 2 weeks. Goal: Working orchestrator → FDA agent → document generation → human review loop.

AI Agent Worker Assignments

Worker	Owned Files / Scope	Acceptance Criteria
Agent-1: API Scaffolder	FastAPI app skeleton, health endpoint, auth stub, Dockerfile	API runs locally, /health returns 200, Docker builds clean
Agent-2: Orchestrator Builder	orchestrator.py, LangGraph graph definition, routing logic	Query routes correctly to FDA agent, state persists across turns
Agent-3: FDA Agent Builder	fda_agent.py, system prompt, tool definitions (RAG lookup, doc draft)	FDA agent returns structured regulatory guidance with citations
Agent-4: RAG Pipeline Builder	ingest.py, chunk strategy, pgvector schema, query interface	20 FDA guidance docs ingested, semantic search returns relevant chunks
Agent-5: Review Queue Builder	review_queue.py, Postgres schema, basic React review UI	Output lands in queue, SME can approve/reject, decision logged
Agent-6: Frontend Builder	React app, chat UI, WebSocket connection, disclaimer display	User can type query, see streaming response, see disclaimer badge
Verifier	All of the above	Full workflow test: query → FDA agent → doc generation → review → delivery

Human Counterpart Task — Sprint 1

Field	Detail
Objective	Procure all technical accounts and API keys needed for Sprint 1 build
Why it matters	No agent can build without live credentials. Blocking dependency.
Steps	1. Create AWS account (or use existing) 2. Create Anthropic API account → generate API key 3. Register domain: latitudemedtech.com (or .ai) 4. Create GitHub org: latitude-medtech 5. Create Clerk or Auth0 account (free tier) 6. Provision RDS Postgres instance (db.t3.micro for dev)
Evidence required	Screenshot of each active account + API key (store in AWS Secrets Manager, NOT shared in chat)
Do NOT	Share API keys in chat or email. Do not use personal accounts for company infrastructure.
Deadline	Before Sprint 1 Day 3 (agents blocked without credentials)

Success criteria	All 5 accounts active, API key stored in Secrets Manager, GitHub org created

8. Verification Gates & Production Blockers

Verification Gate Checklist (per sprint)

- Build / lint / test status: all passing, no critical warnings
- Functional workflow: end-to-end user story works in staging
- Human review gate: no output reaches client without SME approval
- Disclaimer: present on every agent output, correct text, correct placement
- Logging: all agent decisions logged with agent_id, timestamp, inputs, outputs
- Security: no secrets in code, no unauthenticated endpoints, RBAC verified
- Data isolation: client_id scoping verified at DB query level
- Readiness label: output explicitly marked Demo / Alpha / Beta / Production

Production Blockers

All of the following must be resolved before any client-facing deployment:

- Legal entity formed and registered (Latitude MedTech LLC or Corp)
- Terms of Service and Disclaimer reviewed by licensed attorney — explicitly stating outputs are consultant support tools, not legal regulatory advice
- RBAC fully implemented — clients cannot access other clients' data
- Penetration test completed with no critical findings outstanding
- SME human review mandatory for all client-deliverable outputs — no bypass path exists
- ISO-licensed content policy confirmed — no verbatim reproduction of purchased standards
- Data retention and deletion policy documented and implemented
- Incident response runbook written and tested

9. Legal, Security & Compliance Assumptions

- All outputs are labeled as AI-assisted consultant support tools and are NOT a substitute for licensed regulatory affairs advice.
- A licensed regulatory affairs professional (RAC-qualified or equivalent) must review and approve all client-facing deliverables.
- No PHI will be processed in Phase 1–3. HIPAA BAA required before any patient-linked data enters the system.
- ISO standard text (13485, 14971, 62304) will NOT be ingested verbatim. Only Latitude-authored interpretations and procedure templates are used.
- All user data is tenant-isolated. Multi-tenancy security is a Phase 1 requirement, not a Phase 5 nice-to-have.
- The system will maintain a full immutable audit log of all agent decisions and human review actions.
- EU GDPR data subject rights (access, deletion) must be implementable from Phase 5 if EU clients are onboarded.
- Export control review required before any dual-use technology advisory is added to the system.

10. Open Questions

#	Question	Owner	Priority
1	Will Latitude employ licensed RAC consultants in-house or contract them as reviewers?	**Human counterpart**	**Critical**

2	Which ISO standards are licensed for internal use? Needed before RAG ingestion decisions.	**Human counterpart**	**High**
3	Target client segment: startups (pre-submission) or established manufacturers (QMS maintenance)?	**Human counterpart + PM Agent**	**High**
4	Voice feature: internal-only or client-facing from launch?	**Human counterpart**	**Medium**
5	CRM for Sales Agent: HubSpot, Salesforce, or custom?	**Human counterpart**	**Medium**
6	Will the system support notified body interaction facilitation (EU MDR)?	**Orchestrator to research**	**Medium**
7	Pricing model: per-seat SaaS, per-query, or retainer?	**Human counterpart**	**Medium**
8	Multi-language support needed? (EU clients may require DE/FR/IT)	**Defer to Phase 5**	**Low**

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